

# **Diversity of Collembola Morphology as a Bioindicator of Soil Quality Based on Soil Biological Quality Index (QBS-adapt) in Different Agricultural Land Uses**

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## **Abstract**

Soil quality is deteriorating due to intensive soil cultivation for agriculture. The determination of soil quality is primarily done through an approach to the physical and chemical aspect of soil, with biological aspects receiving little to no contribution. Soil Biological Quality Index (QBS) has been applied to evaluate soil quality based on the soil fauna, including Collembola. The objectives of this research were to investigate the diversity of Collembola in different cultivated agriculture soils, to assess soil quality in different cultivated agriculture soils using Collembola diversity as an indicator, and to evaluate the correlations between Collembola diversity and soil's physical and chemical characteristics. Sampling was done using Pit Fall Trap. The samples were observed with a stereo microscope. The diversity index (1,94-2,22) indicated a low diversity level. The evenness index (0,3-0,58) indicated a low evenness level. The dominance index (0,13-0,25) indicated a low dominance level. The score of t Hutcheson test (0,14) showed there was no significant difference between Collembola diversity in each cultivated agriculture soil. The QBS-adapt index values were 11.542 in minimum tillage agriculture; 11.614 in no-tillage agriculture; and 11.766 in secondary forest. PCA analysis revealed the morphological traits that are responsible for the grouping of Collembola morphotypes are furca, ocelli, antennae, and pigmentation. Meanwhile, soil abiotic factors that are responsible for the grouping of Collembola morphotypes are organic matter and total K-content.

*Key words: Collembola, QBS index, soil quality, morphotype, agriculture*

## **INTRODUCTION**

Soil is one of the most important elements, and has become one of the natural resources that is deeply impacted by human activities. In the present industrial era, some human activities threaten to decrease soil quality, such as change of land use, unmanaged disposal of industrial waste, tillage, and agricultural activities. Intense agricultural activities so far have been proven to increase the productivity of fruits and vegetables, but negatively impacted the soil (Bongiorno, 2020). The activity affects the process that happens inside the soil, reducing its function and benefits.

The ecosystems of forest and agricultural lands have different soil structures due to the difference of intensity level in anthropogenic activities. Agricultural lands have a higher level of anthropogenic activities such as manually loosening the soil or by plowing. This practice is called tillage. Studies by Larink (1997) and Pommeresche *et al.* (2014) had proven that the act of harvest decreased the density of Collembola. Trampling on the soil also affects the density

in the soil, decreasing its ability to hold water so that the rate of erosion and water run-off increased (Jagerbrand *et al.*, 2015). Change of land use and increase of agriculture activities impacted the soil by reducing the diversity of soil fauna and loss of soil ecosystem function up to 60% (Diaz *et al.*, 2006; Veresoglou *et al.*, 2015; Singh *et al.*, 2019; Gunstone *et al.*, 2021). Therefore, anthropogenic activities could be assumed as impacting soil arthropods density, diversity, and stability (Goncalves *et al.*, 2020).

Soil invertebrates could be potentially used as one of the means to evaluate soil quality. Soil fauna has a high sensitivity towards anthropogenic activities in the soil surface level (Doran *et al.*, 2000; Juman *et al.*, 2007) hence could be used as an indicator to evaluate soil quality. The diversity and abundance of soil invertebrates are related to soil functions such as decomposition of organic matters (Griffiths *et al.*, 2021), plant development especially in the roots, heavy metal detoxification (Nakatsu *et al.*, 2005), and nutrient mineralization in the soil (Graf *et al.*, 2019; Griffiths *et al.*, 2021).

The evaluation of soil quality using soil fauna as a main indicator is called Qualita Biologica del Suolo (QBS) with arthropods (QBS-ar), Collembola (QBS-c), or an adaptation of the system (QBS-adapt). The QBS index has no need for further identification to species level. The usage of this index is not time-consuming and can be done with simple sampling methods. This index had become the standard of soil quality evaluation protocols in Europe (Menta *et al.*, 2017b) and had been proven effective to evaluate the diversity of microarthropods in forests (Galli *et al.*, 2014; Menta *et al.*, 2017), agricultural lands (Menta *et al.*, 2017), land mine (Madej *et al.*, 2011), and plateau (Agustina *et al.*, 2019; Machado *et al.*, 2019). Although the system has its own advantages, it hasn't been widely spread and commonly applied in Indonesia.

## **MATERIALS AND METHODS**

Soil sampling was carried out at Gedong Songo, Semarang Regency. The process of extraction, sorting, and morphology identification were carried out at the Laboratory of Ecology and Biosystematics, Department of Biology, Diponegoro University. The physical aspects of soil were analyzed at the Laboratory of Soil Mechanics, Department of Civil Engineering, Diponegoro University, and the chemical aspects of soil were analyzed at the Laboratory of Environmental Engineering, Diponegoro University.

Samples of soil fauna were extracted using Pit Fall Trap method. Plots were created using measuring instruments, rope, and pegs in the size of 2x4 meters. Inside each plot, five sampling points were placed diagonally with 1 meter distance in between each point. Soil was dug to the depth of 10 centimeters, and plastic jars filled with unscented detergent and water were placed inside each point. Soil samples in each station were taken to be analyzed further in the laboratory.

The extraction of Collembola was done within 3x24 hours after Pit Fall Trap settlement. The Collembola trapped inside were sorted from the litter and other animals using a pipette. Sorted Collembola were then placed in bottles filled with 95% alcohol to prevent further damages to the sample. Collembola were then observed using stereo microscope RaxVision XPAD-97 and were documented using OptiLab Miconos 4.0.

Data analysis was carried out quantitatively. Collembola were recorded based on observed morphological characteristics and individual abundance at each station. Collembola

is given the codes EP = epigeic (EMI score 0-6), H = hemiedaphic (EMI score 8-12), and E = edaphic (14-20). Collembola community structure was calculated using the Shannon-Wiener diversity index (H'), Simpson dominance index (D), and evenness index (E). The results of calculating the Shannon-Wiener diversity index (H') per sampling location were then compared using the Hutcheson t test to compare the location of differences in diversity values between research locations. Individual abundance per station was tested using IBM SPSS 20 to see differences in abundance between stations. The abundance of individuals per sampling point was used for the one-way ANOVA test, to see whether there were differences in the results of sampling carried out on forest land and agricultural land.

Collembola were identified based on morphological characteristics (Table 1). Collembola individuals with the same EMI value will be classified into the same morphotype. The sum of the EMI values is referred to as the final EMI value of morphotype n. This value is used further to calculate the QBS index. Morphological characteristics were observed and calculated according to the Ecomorphological Index (EMI).

Table 1. Morphological characteristics of Collembola based on EMI (Machado *et al.*, 2019)

| Morphology          | Characteristics                    | EMI Value |
|---------------------|------------------------------------|-----------|
| Number of ocelli    | Absent                             | 0         |
|                     | Present                            | 4         |
| Length of antennae  | Antennae length > body length      | 0         |
|                     | Antennae length > 0.5x body length | 2         |
|                     | Antennae length < 0.5x body length | 4         |
| Furca               | Present, well developed            | 0         |
|                     | Present, reduced                   | 2         |
|                     | Absent                             | 4         |
| Hairs and/or scales | Present                            | 0         |
|                     | Absent                             | 4         |
| Pigmentation        | Colored with patterns              | 0         |
|                     | Colored, uneven                    | 2         |
|                     | Transparent                        | 4         |

The EMI value is used to calculate the QBS index. The QBS index value per sampling location is used to determine the comparison of soil quality. The QBS-adapt index on untreated soil can reach a value of 23,400, natural forests can reach a value of 66,442 (Machado et al., 2019). QBS index based on Machado et al. (2019) is calculated using the formula as seen below

$$QBS = \sum_{i=1}^n (EMI \text{ value} \times \text{abundance})$$

with QBS = QBS value/station

EMI = EMI final value/morphotype

Abundance = total of individual in each morphotype

The soil chemical factors analyzed are pH, organic matter (%), N (%), P (%), and K (%), while the soil physical factors analyzed are temperature (°C), humidity (%), and soil

texture. The number of Collembola morphotypes obtained, the morphological characteristics observed, as well as the physical and chemical properties of the soil on forest and agricultural land were used for PCA (Principal Component Analysis) analysis to see the influence of soil physicochemical factors and morphological characters on the grouping of Collembola morphotypes. PCA analysis was carried out using MVSP 3.2 software.

## RESULTS

### Collembola Diversity in Different Agricultural Lands

Based on the observation results (Table 2), there were a total of 22 Collembola morphotypes found in agricultural land with minimum tillage, no tillage, and secondary forest. Based on the observations, the hemiedaphic Collembola morphotype (14 morphotypes) had the highest number of morphotypes compared to the epigeic morphotypes (7 morphotypes) and edaphic (1 morphotype). Morphotype grouping was carried out based on the similarity of the number of eco-morphological index (EMI) values, namely Collembola epigeic (EMI score 0-6), hemiedaphic (EMI score 8-12), and edaphic (14-20) referring to the QBS-adapt system (Machado *et al.*, 2019).

Table 2. Collembola morphology classification

| Characteristics |          |       |                |              | EMI | Ecomorphological Group | Morphotype |
|-----------------|----------|-------|----------------|--------------|-----|------------------------|------------|
| Ocelli          | Antennae | Furca | Hairs / scales | Pigmentation |     |                        |            |
| 0               | 2        | 0     | 0              | 0            | 2   | Epigeic                | EP1        |
| 0               | 2        | 0     | 0              | 2            | 4   | Epigeic                | EP2        |
| 0               | 4        | 0     | 0              | 0            | 4   | Epigeic                | EP3        |
| 0               | 2        | 0     | 0              | 4            | 6   | Epigeic                | EP4        |
| 0               | 2        | 4     | 0              | 0            | 6   | Epigeic                | EP5        |
| 0               | 4        | 0     | 0              | 2            | 6   | Epigeic                | EP6        |
| 4               | 2        | 0     | 0              | 0            | 6   | Epigeic                | EP7        |
| 0               | 4        | 0     | 0              | 4            | 8   | Hemiedaphic            | H1         |
| 0               | 4        | 4     | 0              | 0            | 8   | Hemiedaphic            | H2         |
| 4               | 0        | 0     | 0              | 4            | 8   | Hemiedaphic            | H3         |
| 4               | 2        | 0     | 0              | 2            | 8   | Hemiedaphic            | H4         |
| 4               | 4        | 0     | 0              | 0            | 8   | Hemiedaphic            | H5         |
| 0               | 2        | 4     | 0              | 4            | 10  | Hemiedaphic            | H6         |
| 0               | 4        | 4     | 0              | 2            | 10  | Hemiedaphic            | H7         |
| 4               | 2        | 0     | 0              | 4            | 10  | Hemiedaphic            | H8         |
| 4               | 2        | 4     | 0              | 0            | 10  | Hemiedaphic            | H9         |
| 4               | 4        | 0     | 0              | 2            | 10  | Hemiedaphic            | H10        |
| 0               | 4        | 4     | 0              | 4            | 12  | Hemiedaphic            | H11        |
| 4               | 2        | 4     | 0              | 2            | 12  | Hemiedaphic            | H12        |
| 4               | 4        | 0     | 0              | 4            | 12  | Hemiedaphic            | H13        |
| 4               | 4        | 4     | 0              | 0            | 12  | Hemiedaphic            | H14        |
| 4               | 4        | 4     | 0              | 2            | 14  | Edaphic                | E1         |

EMI scores 0 - 6 = edaphic, EMI scores 8 - 12 = hemiedaphic, EMI scores 14 - 20 = edaphic. Scoring values referred to Table 1.

The total of Collembola found at the three sampling stations was 2,378 individuals. The highest abundance of Collembola was found in no tillage land use with a total of 1,154 individuals. The smallest abundance of Collembola was found in the secondary forest with a total of 460 individuals. Overall Collembola abundance data can be seen in Table 3.

Table 3. Abundance of Collembola in minimum tillage, no tillage, and secondary forest

| Ecomorphological Group           | Number of Individual |            |                  | Total |
|----------------------------------|----------------------|------------|------------------|-------|
|                                  | Minimum Tillage      | No Tillage | Secondary Forest |       |
| EP1                              | 123                  | 250        | 210              | 583   |
| EP2                              | 109                  | 54         | 68               | 231   |
| EP3                              | 137                  | 540        | 44               | 721   |
| EP4                              | 33                   | 14         | 8                | 55    |
| EP5                              | 0                    | 0          | 1                | 1     |
| EP6                              | 14                   | 62         | 20               | 96    |
| EP7                              | 44                   | 29         | 20               | 93    |
| EP Total                         |                      |            |                  | 1,780 |
| H1                               | 4                    | 5          | 3                | 12    |
| H2                               | 2                    | 0          | 0                | 2     |
| H3                               | 1                    | 0          | 0                | 1     |
| H4                               | 122                  | 72         | 7                | 201   |
| H5                               | 26                   | 42         | 14               | 82    |
| H6                               | 0                    | 0          | 14               | 14    |
| H7                               | 0                    | 0          | 3                | 3     |
| H8                               | 105                  | 14         | 19               | 138   |
| H9                               | 2                    | 5          | 2                | 9     |
| H10                              | 5                    | 28         | 2                | 35    |
| H11                              | 0                    | 0          | 20               | 20    |
| H12                              | 0                    | 8          | 0                | 8     |
| H13                              | 6                    | 9          | 3                | 18    |
| H14                              | 26                   | 15         | 2                | 43    |
| H Total                          |                      |            |                  | 598   |
| E1                               | 5                    | 7          | 0                | 12    |
| Total of Individual (EP + H + E) |                      |            |                  | 2,378 |

EP = epigeic, H = hemiedaphic, E = edaphic. The numbers followed referred to the order of morphotypes

The results of the diversity, dominance, and evenness indices of Collembola in minimum tillage, no tillage, and secondary forest land can be seen in Table 4.

Table 4. Diversity, dominance, and evenness indices values of Collembola

| Station          | Number of Individual | Number of Morphotype | Diversity Index | Dominance Index | Evenness Index |
|------------------|----------------------|----------------------|-----------------|-----------------|----------------|
| Minimum Tillage  | 764                  | 17                   | 2,22            | 0,13            | 0,54           |
| No Tillage       | 1.154                | 16                   | 1,78            | 0,28            | 0,37           |
| Secondary Forest | 460                  | 18                   | 1,94            | 0,25            | 0,38           |

The difference of Collembola abundance at minimum tillage, no tillage, and secondary forest land can be seen in Table 5.

Table 5. *p* values of Hutcheson test in the abundance of Collembola at minimum tillage, no tillage, and secondary forest

| Station          | Minimum Tillage       | No Tillage            | Secondary Forest     |
|------------------|-----------------------|-----------------------|----------------------|
| Minimum Tillage  |                       | 8,1249 <sup>-21</sup> | 1,4574 <sup>-5</sup> |
| No Tillage       | 8,1249 <sup>-21</sup> |                       | 0,02306              |
| Secondary Forest | 1,4574 <sup>-5</sup>  | 0,02306               |                      |

*p* value < 0,05 expressed a significant difference between stations

ANOVA was performed to comprehend the difference of the average values of Collembola abundance in three stations. The significance value can be seen in Table 6.

Table 6. One-way ANOVA test values of the average Collembola abundance in minimum tillage, no tillage, and secondary forest land

|                  | Average                  | F    | Sig. |
|------------------|--------------------------|------|------|
| Minimum Tillage  | 50,93±25,39 <sup>a</sup> | 2,79 | 0,14 |
| No Tillage       | 76,93±51,35 <sup>a</sup> |      |      |
| Secondary Forest | 30,66±5,20 <sup>a</sup>  |      |      |

Average±standard of deviation. The letter *a* signified that there was no significant difference of the average values in three stations

### Soil Quality Analysis Based On Collembola Diversity

In regards to the QBS-adapt index, the calculations were carried out based on the modification of QBS-ar index by Machado *et al.* (2019). The QBS-adapt index results for minimum tillage, no tillage, and secondary forest land can be seen in Table 7.

Table 7. QBS-adapt index calculations in minimum tillage, no tillage, and secondary forest land

| Morphotype         | EMI | Abundance | EMI x<br>abundance | Station            |            |                     |
|--------------------|-----|-----------|--------------------|--------------------|------------|---------------------|
|                    |     |           |                    | Minimum<br>Tillage | No Tillage | Secondary<br>Forest |
| EP1                | 2   | 583       | 1.166              | 1.166              | 1.166      | 1.166               |
| EP2                | 4   | 231       | 924                | 924                | 924        | 924                 |
| EP3                | 4   | 721       | 2.884              | 2.884              | 2.884      | 2.884               |
| EP4                | 6   | 55        | 330                | 330                | 330        | 330                 |
| EP5                | 6   | 1         | 6                  | -                  | -          | 6                   |
| EP6                | 6   | 96        | 576                | 576                | 576        | 576                 |
| EP7                | 6   | 93        | 558                | 558                | 558        | 558                 |
| H1                 | 8   | 12        | 96                 | 96                 | 96         | 96                  |
| H2                 | 8   | 2         | 16                 | 16                 | -          | -                   |
| H3                 | 8   | 1         | 8                  | 8                  | -          | -                   |
| H4                 | 8   | 201       | 1.608              | 1.608              | 1.608      | 1.608               |
| H5                 | 8   | 82        | 656                | 656                | 656        | 656                 |
| H6                 | 10  | 14        | 140                | -                  | -          | 140                 |
| H7                 | 10  | 3         | 30                 | -                  | -          | 30                  |
| H8                 | 10  | 138       | 1.380              | 1.380              | 1.380      | 1.380               |
| H9                 | 10  | 9         | 90                 | 90                 | 90         | 90                  |
| H10                | 10  | 35        | 350                | 350                | 350        | 350                 |
| H11                | 12  | 20        | 240                | -                  | -          | 240                 |
| H12                | 12  | 8         | 96                 | -                  | 96         | -                   |
| H13                | 12  | 18        | 216                | 216                | 216        | 216                 |
| H14                | 12  | 43        | 516                | 516                | 516        | 516                 |
| E1                 | 14  | 12        | 168                | 168                | 168        | -                   |
| Total              |     | 2.378     |                    |                    |            |                     |
| QBS-adapt<br>Index |     |           |                    | 11.542             | 11.614     | 11.766              |

### Correlation Between Collembola Morphotype Diversity and Soil Abiotic Properties

To complement the evaluation of soil quality with QBS-adapt index, the results of the physical and chemical properties analysis of soils sampled were shown in Table 8.

Table 8. Physical and chemical properties of soils at minimum tillage, no tillage, and secondary forest land use

| Station    | Temperature | Moisture | pH | Total N | Total P | Total K | Organic Matter |
|------------|-------------|----------|----|---------|---------|---------|----------------|
|            | (°C)        | (%)      |    | (%)     | (%)     | (%)     | (%)            |
| Minimum    | 22          | 66       | 8  | 0,638   | 0,189   | 0,006   | 42,1           |
| Tillage    |             |          |    |         |         |         |                |
| No Tillage | 21          | 50       | 8  | 0,79    | 0,116   | 0,011   | 11,13          |
| Secondary  | 19          | 56       | 8  | 0,694   | 0,056   | 0,007   | 24,06          |
| Forest     |             |          |    |         |         |         |                |

Furthermore, PCA analysis was carried out to pinpoint the most contributing aspects of the biological (morphology), physical, and chemical properties to the grouping of Collembola morphotypes in three stations. The results of PCA analysis were shown in Table 9.

Table 9. *Eigenvalues* of each morphological characteristics and physical-chemical properties to the diversity of Collembola morphotypes

|                    | PC1    | PC2    | PC3    |
|--------------------|--------|--------|--------|
| <i>Eigenvalues</i> | 1,521  | 0,285  | 0,261  |
| % Diversity        | 62,233 | 11,646 | 10,691 |
| % Cumulative       | 62,233 | 73,879 | 84,570 |
| Diversity          |        |        |        |

## DISCUSSIONS

### Collembola Diversity in Different Agricultural Lands

The highest abundance of the Collembola eco-morphological group is found in the epigeic group, or Collembola that live on the soil surface. As much as 1.780 individuals of epigeic Collembola were found and made up around 74.9% of the total abundance of Collembola. The smallest abundance of eco-morphological groups is found in the edaphic group, or Collembola that live below the soil surface. This can be caused by the method used, the Pit Fall Trap method, which is a method used to catch arthropods that live on the surface level of soil. This is in accordance with the opinion of Jaya *et al.* (2018) which states that the Pit Fall Trap method functions to trap small animals such as insects that have high mobility on the ground surface.

The calculation results of the Shannon-Wiener diversity index ( $H'$ ) (Table 4) show a value of 2.22 at minimum tillage station; value 1.78 at no tillage station; and a value of 1.94 for secondary forest station. This shows that the three stations have moderate diversity (Odum, 1993; Ulfah *et al.*, 2019). However, the minimum tillage station has the highest diversity index value compared to the other two stations. The high diversity index value at the minimum tillage station can be caused by the high number of individuals per morphotype, as well as the even distribution of individuals. This is supported by Hubalek (2000) and Mulya *et al.* (2021) which states that the diversity index is calculated by the number of species and the number of individuals, hence the differences between the two will produce significantly different results.

The results of Hutcheson's t test calculations (Hutcheson, 1970) (Table 5) were carried out to determine the differences in the Shannon-Wiener diversity index ( $H'$ ) by comparing the diversity index values of two stations. According to Printer *et al.* (2019), Hutcheson's t test hypothesis is  $H_0$  there is no difference between two habitats based on the Shannon-Wiener diversity index, and  $H_1$  there is a difference between two habitats based on the Shannon-Wiener diversity index. The agricultural land use with minimum tillage and no tillage have a  $p$  value of  $8,1249^{-21}$  ( $p < 0,05$ ), the agricultural land with no tillage and secondary forest land use have a  $p$  value of 0,02306 ( $p < 0,05$ ), while the agricultural land with minimum tillage and secondary forest land use has a  $p$  value of  $1,4574^{-5}$  ( $p < 0,05$ ). Statistically, there is the highest difference value between no tillage and secondary forest compared to other land use systems, meaning that there is a significant difference in the diversity of Collembola population in no tillage and secondary forest.

The results of the calculation of the evenness index ( $E$ ) (Table 4) were carried out to determine the evenness of population distribution in the sampling area. The minimum tillage



land use has an evenness index (E) value of 0,54; no tillage land use has an evenness index (E) value of 0,37; while secondary forest has an evenness index (E) value of 0,38. The minimum tillage land use has the highest Collembola population evenness compared to the other two stations. This means that the Collembola population at the minimum tillage land use is the most even and stable compared to no tillage and secondary forest. According to Krebs (1989) and Ulfah et al. (2019), the category of evenness index (E) value of less than 0,5 is said to be a very unstable and evenly distributed community, while the evenness index (E) value between 0,5 and 0,75 is included in the category of unstable community because of the low evenness. The smaller the evenness index value, the smaller the population uniformity.

The dominance index (D) is used to determine the dominance of a species in the sampling area. The results of the dominance index (D) (Table 4) on the minimum tillage land use has a dominance index (D) value of 0.13; no tillage land use has a dominance index (D) value of 0,28; while secondary forest has a dominance index (D) value of 0,25. Based on these values, the three stations have a low level of dominance (Odum, 1993; Ulfah *et al.*, 2019). However, the agricultural land use with no tillage has the highest dominance value compared to the other two stations.

Data that has been proven to be normally distributed and homogeneous is then used for the ANOVA test. The ANOVA test result showed a value of 0,14 (Table 6). This value can be interpreted as there is no significant difference in the average number of samples at the minimum tillage land use, no tillage land use, and secondary forest. This is in accordance with the opinion of Muhson (2016), which stated that if the  $p$  value is greater than the significance value ( $p > 0.05$ ) then it could be said that there is no significant difference in the average value of the abundance of Collembola individuals at the agricultural land use with minimum tillage, no tillage, and secondary forest.

### **Soil Quality Analysis Based On Collembola Diversity**

Calculation of the QBS-adapt index at the three stations produced a value of 11.542 for minimum tillage land use, 11.614 for no tillage, and 11.766 for secondary forest (Table 7). Secondary forest has the highest QBS-adapt value (11.766) compared to minimum tillage (11.542) and no tillage (11.614). This value indicates that secondary forest land has the best soil quality based on the evaluation of Collembola diversity. According to Parisi *et al.* (2001) and Machado et al. (2019), the better the environmental quality, the higher the number of microarthropods that will adapt to the soil.

The difference on QBS-adapt index values between two different tillage systems in the agricultural land is expected. No tillage (11.614) has a higher QBS-adapt index value than minimum tillage (11.542). This could be possible because land cultivation practices damage the Collembola population structure in the soil, hence the number and diversity of Collembola is less at the minimum tillage than at the no tillage agricultural land. Tilling the soil on agricultural land is a strategy for loosening the soil with or without tools, so that the land can be planted with various kinds of agricultural commodities. However, land cultivation practices actually disrupt the soil fauna community within it. This is supported by Wardle (1995) and Coulibaly *et al.* (2017) who stated that the abundance of Collembola increased from land management with conventional tillage to land without tillage due to the absence of

environmental pressure. According to Gardi *et al.* (2008) and Machado *et al.* (2019), land cultivated using agricultural equipment had a lower QBS index value than forest and farm.

### **Correlation Between Collembola Morphotype Diversity and Soil Abiotic Properties**

Based on the research results (Table 8), the three stations showed the same pH value (8). The soil on the minimum tillage land use had a temperature of 22°C; humidity value of 66%; and the soil was composed of 47.44% fine sand, 40.2% silt and 0% clay. The soil on no tillage had a soil temperature of 21°C; humidity of 50%; and the soil was composed of 62.90% silt; 22.06% fine sand; and 2.00% clay. Soil in the secondary forest had a temperature of 19°C; humidity of 56%; and the soil was composed of 53.24% silt; 37.74% fine sand; and 0.00% clay. The composition percentage was used in calculating the soil texture triangle. The result showed that the soil structure at the three stations were silt loam or sandy clay soils. The minimum tillage land use had a total nitrogen (N) concentration of 0.64%; total phosphorus (P) concentration of 0.19%; total potassium (K) concentration of 0.01%; and organic matter 42.1%. The no-tillage land use had a total nitrogen (N) concentration of 0.79%; total phosphorus (P) concentration of 0.12%; total potassium (K) concentration of 0.01%; and organic matter of 11.13%. Secondary forest land had a total nitrogen (N) concentration of 0.69%; total phosphorus (P) concentration of 0.06%; total potassium (K) concentration of 0.01%; and organic matter of 24.06%.

Based on the results of PCA analysis (Table 9), Principal Component 1 (PC1), PC2, and PC3 showed a cumulative diversity percentage value of 62.23%; 73.88%; and 84.57% consecutively. This meant that PC1, PC2, and PC3 were the three main components to explain the grouping of Collembola morphotypes based on soil morphology and abiotic properties. This is supported by Filho *et al.* (2021) which stated that the main components that can explain data variations have at least a cumulative diversity that reaches 80%.

Principal component 1 (PC1) showed the highest *eigenvalue* (1.52) with a diversity percentage of 62.23%. Furca played a main role in PC1 diversity, along with temperature, humidity, total nitrogen (N), total phosphorus (P), total potassium (K), and organic matter. Principal component 2 (PC2) showed an *eigenvalue* of 0.29; diversity percentage of 11.65%; and the cumulative diversity percentage of 73.88%. Morphological characters that played a role in PC2 diversity were ocelli, antennae, and pigmentation. Principal component 3 (PC3) showed an *eigenvalue* of 0.26; diversity percentage of 10.69%; and the cumulative diversity percentage of 84.57%. Morphological characters that played a role in PC3 diversity were ocelli, antennae, furca, and pigmentation.

A scatter plot is a graph that shows the relationship between variables. The values for these variables appeared along the x-axis and y-axis, while the individuals in the data appeared as points on the graph (Moore *et al.*, 2013; Bergstrom *et al.*, 2018). According to ter Braak *et al.* (1995), the direction of the arrow indicated the species with the greatest abundance, while the length of the line indicated the level of change in that direction. The results of scatter plot analysis (Figure 1) showed the grouping of Collembola morphotypes. Eco-morphotype group 1 consisted of sub-group 1 (morphotype E1) and sub-group 2 (morphotypes H2, H3, H6, H11, H12, and EP5). Eco-morphotype group 2 consisted of sub-group 3 (morphotypes EP1, EP3, EP4, EP7, H5, H8, H13, and H14) and sub-group 4 (H1 and H9). The longest vector lines were C, L, and K which showed that the furca, total potassium, and organic matter were the main

properties that were responsible for the grouping of Collembola morphotypes into three huge eco-morphotype groups.

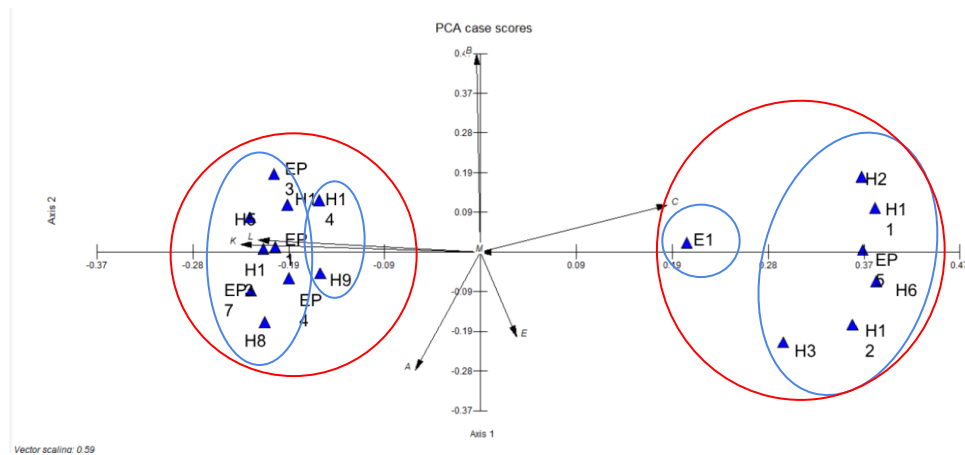


Figure 1. Scatter plot analysis of the Collembola morphotype grouping based on the morphological characteristics and soil abiotic properties

In the scatter plot analysis, there were three main characteristics that contribute to the grouping of Collembola morphotypes, namely the properties C (furca), K (total potassium), and L (organic matter). Eco-morphotype group 1 was grouped based on similarity to the trait C (furca). Furca is an organ used by Collembola to move, supporting leg function (Pommeresche et al., 2014). The members of eco-morphotype group 1 (E1, H2, H3, H6, H11, H12, EP5) don't possess a furca in their morphology. This is related to their eco-morphological form, namely edaphic and hemiedaphic, which live below the surface of the land with limited space for movement. According to Hopkins (1997) and Bellini et al. (2023), Collembola species that live on the surface of the soil have furca and tenaculum that are reduced or even absent altogether. This is due to Collembola's limited space for movement.

Organic matter and total potassium contributed to the grouping of eco-morphotype group 2 (EP1, EP3, EP4, EP7, H1, H5, H8, H9, H13, and H14). Eco-morphotype group 2 can be found at the minimum tillage land use, no tillage, and secondary forest with more varied environmental conditions, namely soil with a total potassium percentage of 0.006 - 0.01%; while the percentage value of organic material was 11.13 - 42.1%. The lines showing the properties of total potassium and organic matter were shown as adjacent to the same direction. This indicated that the two variables are positively correlated, in accordance with the opinion of ter Braak *et al.* (1995). Two variables that have a positive correlation mean that if the total potassium value increased, the organic material value would also increase, and vice versa. High potassium values come from agricultural practices that use manure, compost, and plant residues that fall to the ground (Sharma *et al.*, 2009; Kaur *et al.*, 2019). The act of increasing the potassium value will also increase the existing organic matter in the soil. This is supported by Baruah *et al.* (1991), Guzel *et al.* (2006), and Ngwe *et al.* (2012) which stated that potassium material in soil is positively correlated with organic matter.

## CONCLUSIONS

The Shannon-Wiener diversity index ( $H'$ ) showed a value of 2.22 in minimum tillage land use; 1.78 in no tillage; and 1.96 in secondary forest which indicated moderate diversity of the Collembola population. The dominance index ( $D$ ) showed a value of 0.13 in minimum tillage land use; 0.28 in no tillage; and 0.25 in secondary forest which indicated low dominance of Collembola morphotypes. The evenness index ( $E$ ) showed a value of 0.54 in minimum tillage land use; 0.37 in no tillage; and 0.38 in secondary forest which indicated low evenness of Collembola distribution. The ANOVA test showed a value of 0.14 which stated that there was no significant difference in the average abundance of Collembola individuals at the three stations. The results of the QBS-adapt index indicated that secondary forest has the best soil quality compared to the other two stations, as indicated by the QBS index value for minimum tillage land use is 11.542, no tillage is 11.614, and secondary forest is 11.766. Based on PCA analysis, the morphological characteristics that have the most influence on the grouping of Collembola morphotypes are furca, ocelli, antennae, and pigmentation. The abiotic factors that have the most influence on the grouping of Collembola morphotypes are total potassium and organic matter.

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